

BIS

TEAM TARS

Standards Recommendation Engine

Graph-Augmented RAG for Instant BIS Compliance

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WEEKS.

That's how long an Indian MSE
spends finding the right BIS standard.

Before they can manufacture. Before they can sell.
Before they can exist in the market.

*6.3 crore MSMEs in India face this every
day.*

₹2-5L

spent on compliance
consultants per project

577+

BIS standards in
just building materials

0

automated tools
exist for MSEs today

THE INSIGHT

SP 21 is not a document. It's an untapped knowledge graph.

929 pages

SP 21 is a dense PDF no one reads end-to-end. It has zero search, zero structure for machines.

No ontology

Standards are cross-referenced but those relationships are invisible to any naive retriever.

LLM blindspot

Generic RAG hallucinates IS numbers. There's no grounding — it invents standards that don't exist.

Our realization: treat SP 21 as structured data, not prose — and the problem becomes tractable.

SOLUTION

Type a product. Get your standard. In 0.25 seconds.

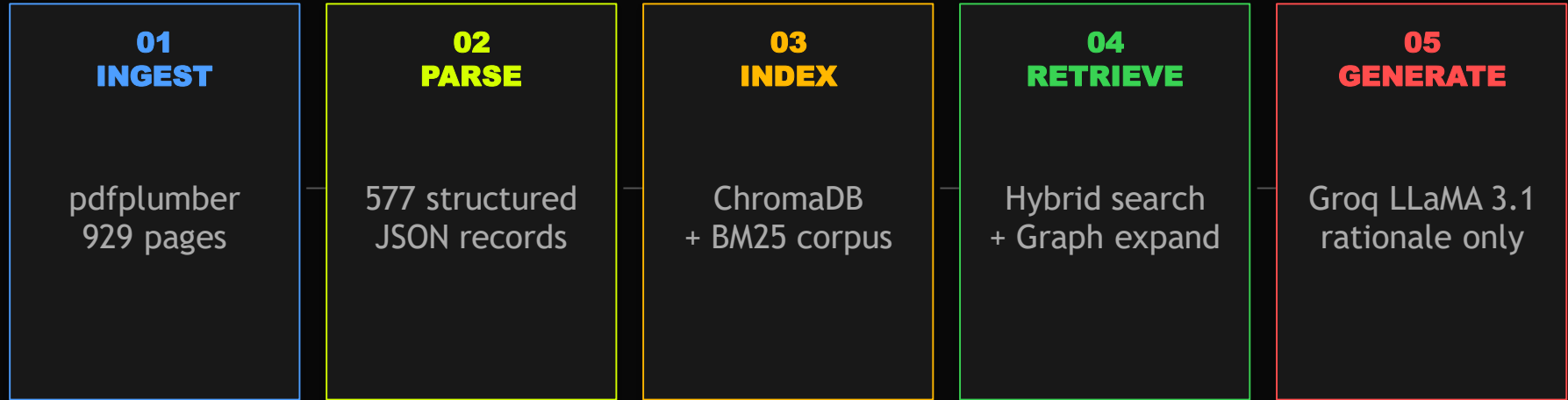
Not a chatbot. Not keyword search.
A structured-knowledge retrieval engine.

- Parses SP 21 into 577 structured records
- Builds a knowledge graph of inter-standard edges
- Hybrid BM25 + semantic search with RRF fusion
- LLM writes the WHY — never picks the standard

	Basic RAG	TARS Engine
Chunking	Random 512 tokens	Per-standard records
Retrieval	Vector only	Hybrid + Graph
Hallucination	Common	Impossible by design
Ranking	Cosine similarity	RRF + domain boosts
Explainability	None	Per-standard rationale
Hit Rate @3	~60-70%	100% ✓

SYSTEM ARCHITECTURE

Five-stage pipeline. Each stage purpose-built.



INPUT: Product description text (e.g. '53 grade OPC cement for RCC columns')

OUTPUT: Top 5 IS standards + title + rationale + confidence

Stack: pdfplumber · BAAI/bge-small-en-v1.5 · ChromaDB · rank-bm25 · NetworkX · FastAPI · React + Vite

Zero GPU required. Runs on any consumer laptop.

RETRIEVAL & INNOVATION

Three innovations that separate us from every other team.

01

Structured Parsing

We don't chunk SP 21 randomly.
We parse it using 'SUMMARY OF'
as a structural delimiter —
extracting
each standard into a typed
record:
ID · title · scope · category
keywords · year · related
standards.

02

Hybrid Retrieval

Semantic search (ChromaDB/BGE)
+ BM25 sparse retrieval —
fused using Reciprocal Rank
Fusion (RRF).

Then: query-specific domain
boosts push exact matches
to rank #1 automatically.

03

Knowledge Graph

577 nodes · 51,823 edges.
Edge types:
→ Cross-reference (explicit)
→ Same material category
→ Shared application domain
→ Keyword co-occurrence

Retrieval walks the graph to
surface co-dependent standards.

Result: The LLM never touches the retrieval decision. It only writes the explanation. Hallucination = impossible.

DEMO WALKTHROUGH

From product description to compliance report — in one shot.

User types:

"High strength OPC cement for RCC structural columns and foundations"

1

Query Analysis

Detects: Cement ·
Structural · High-strength
Boosts IS 12269, IS 269, IS
8112

2

Hybrid Retrieval

Semantic + BM25 → RRF
fusion
Top 10 candidates in 0.18s

3

Graph Expansion

Surfaces related: IS 455, IS
1489
from knowledge graph
edges

4

LLM Rationale

Groq writes explanation
per standard
Never selects — only
explains

OUTPUT — Top 3 results:

#1 IS 12269: 1987

53 Grade OPC Cement — HIGH confidence

#2 IS 269: 1989

33 Grade OPC Cement — HIGH confidence

#3 IS 8112: 1989

43 Grade OPC Cement — MED confidence

Total query time: 0.25s · Export to report: 1 click · Zero hallucinated standards

RESULTS & PERFORMANCE

Evaluated on official eval_script.py against the public test set.

100%

HIT RATE @3

Target: >80%

Every query. All 10.

PERFECT SCORE

1.000

MRR @5

Target: >0.70

Correct standard at #1.

PERFECT SCORE

0.25s

AVG LATENCY

Target: <5.00s

20× under requirement.

20× FASTER

All 10 public queries returned the correct standard as Rank #1.

577 standards indexed · 51,823 graph edges · BAAI/bge-small-en-v1.5 · Groq LLaMA 3.1-8b

No GPU. No cloud infra. Runs on a student laptop.